

Optoelectronic Biointerfaces for Ex Vivo Retina Stimulation

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Background: Degenerative retinal diseases such as age-related macular degeneration and retinitis pigmentosa result in photoreceptor loss with disruption in native phototransduction while largely preserving inner retinal circuitry. Artificial phototransduction strategies propose stimulation of surviving retinal neurons. Photovoltaic biointerfaces offer a wireless approach by converting incident light into electrical currents without external power sources.

Methods: We developed quantum dot integrated devices designed to generate pseudocapacitive currents under near infrared light illumination. The multilayer architecture was optimized for efficient photocurrent generation. Biocompatibility of the photovoltaic interface was first assessed in primary hippocampal neuron cultures through viability assays and neuronal marker analysis. Device functionality was evaluated using *ex vivo* retinal explants isolated from healthy and degenerated animals recorded on multi-electrode array (MEA) systems. Illumination (880 nm, 5Hz, 20ms ON, 180ms OFF cycles) was applied to the entire device surface, and neuronal responses were quantified through changes in active electrode number and firing dynamics.

Results: NIR illumination induced robust light-triggered modulation of retinal activity. Photovoltaic stimulation significantly increased the number of active electrodes compared to baseline recordings and altered firing dynamics across the network, indicating effective recruitment of retinal neurons.

Conclusion: Our findings demonstrate that a quantum dot-based photovoltaic interface can effectively modulate retinal network activity under NIR light illumination. This wireless, optoelectronic strategy represents a promising platform for next-generation subretinal prosthetic systems targeting photoreceptor degeneration.

Keywords: Retinal prosthesis; vision restoration; bioelectronic interface; NIR light stimulation